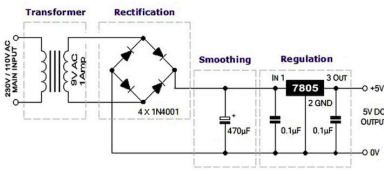




5 volt DC Power Supply using 7805



Formulas needed: $V = iR$

$$P = iV$$

$$V_{rms} = \frac{V_{pk}}{\sqrt{2}}$$

$$\Delta V = \frac{I}{2 \times 5 \times C}$$

$$P_{heat} = (V_{in} - V_{out}) \times I$$

Building in LTSpice:

Transformer:

Diagram has a Transformer, 9V AC, 1Amp

RMS Value is 9V, converting to peak sine value

$$V_{pk} = V_{rms} \cdot \sqrt{2}$$

$$V_{pk} = 9 \cdot \sqrt{2}$$

$$V_{pk} = 12.72792206 \text{ V}$$

Notes: RMS behaves like DC and is given

for power calculations. Also to see how

hot a resistor gets.

• Peak sine wave is for voltage limit questions,

like whether a capacitor's 25V rating is safe, etc.

The bridge rectifier:

Idea: A diode is a one-way valve for current: it

connects from anode to cathode (the way the arrow points)

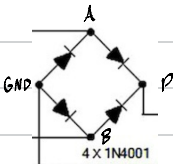
and blocks the other way. There are four diodes so on

both halves of the swing, current flows through the load

the same way.

$$V_{out, peak} = V_{peak} - [2 \times V_d], \text{ (} V_d \text{ is voltage dropped across one conducting diode)}$$

2 Voltage drops because current passes through two diodes on each half cycle.

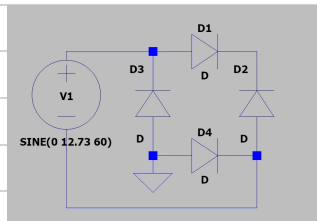


$$D_1: A \rightarrow P$$

$$D_2: B \rightarrow P$$

$$D_3: GND \rightarrow A$$

$$D_4: GND \rightarrow B$$



Capacitor and load:

Bridge output is a series of bumps that fall to 0V between peaks,

The capacitor stores charge at each peak and supplies the load between them.

$$\Delta V = \frac{I}{2 \times f \times C}$$

$$\Delta V = \frac{0.5}{2 \times 60 \times 0.0022}$$

$$\Delta V = 1.89 \text{ V}$$

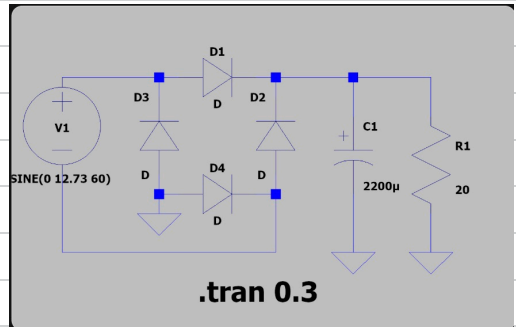
Choosing load resistor:

DC voltage will be roughly 10 V, I want about 0.5 A,

$$R = \frac{V}{I}$$

$$R = \frac{10}{0.5}$$

$$R = 20 \Omega$$



- Multiple grounds are fine, they all represent the same node (0V)

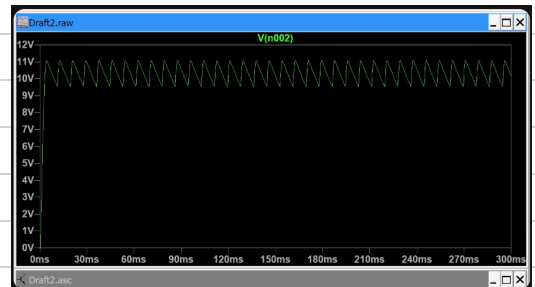
After running simulation:

Quantity	Calculated	Simulated (from plot)	% error
Peak	11.13 V	= 11.05 V	-0.7%
Valley (minimum)	9.24 V	= 9.55 V	+3.4%
Ripple (peak - valley)	1.89 V	= 1.5 V	-21%

Percent error: Peak: $\frac{11.05 - 11.13}{11.13} \times 100\% = -0.7\%$

Valley (min): $\frac{9.24 - 9.55}{9.55} \times 100\% = -3.4\%$

Ripple (max-min): $\frac{1.5 - 1.89}{1.89} \times 100\% = -21\%$



Why is ripple smaller than predicted?

The formula $\Delta V = \frac{I}{2 \times f \times C}$ assumed the capacitor discharges

for the whole 8.33 ms between peaks. In the simulation, it only discharges until the next peak brings the diodes back on, which happens sooner.

Check:

• Average rail Voltage = $\frac{11.05 + 9.55}{2} = 10.3 \text{ V}$, $I = \frac{10.3}{20} = 0.515 \text{ A}$

• Discharge time implied by plot: $t = \frac{\Delta V \times C}{I} = \frac{1.5 \times 0.0022}{0.515} = 6.4 \text{ ms}$

So diodes conduct for about $8.33 - 6.4 = 1.9 \text{ ms}$ of every half-cycle.

My formula is a conservative upper bound, so designing from it gives extra margin. The real circuit should land near the simulation.

Design check: dropout margin

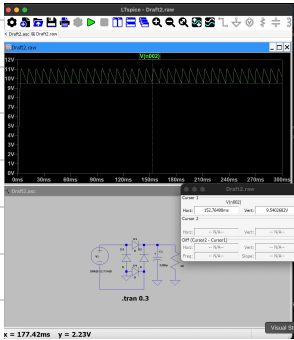
• Voltage = 9.55 V, 7805 needs 7 V, margin is 2.55 V

• With 10% sag in adapter: peak $\approx 8.1 \times 1.414 - 1.68 = 9.77 \text{ V}$, The Valley is $9.77 - 1.5 = 8.27 \text{ V}$, still above 7 V ✓

Implied diode drop

$\frac{12.73 - 11.05}{2} = 0.84 \text{ V}$ per diode at current spikes. That's the library 1N4007 model, my real diodes will differ.

Reading graph using cursors:



Valley:

Valley (simulated) = 9.54 V

Error vs calculated: $\frac{9.54 - 9.24}{9.24} \times 100\% = 3.2\%$

Margin above 7805's requirement: $9.54 - 7 = 2.54 \text{ V}$

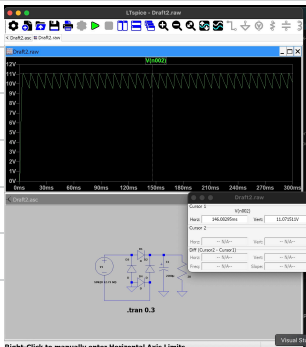
Quantity	Calculated	Simulated (cursor)	% error
Peak	11.13 V	11.07 V	-0.5%
Valley	9.24 V	9.54 V	+3.2%
Ripple	1.89 V	1.53 V	-19%

% error:

Peak: $\frac{11.07 - 11.13}{11.13} \times 100\% = -0.5\%$

Valley: $\frac{9.54 - 9.24}{9.24} \times 100\% = 3.2\%$

Ripple: $\frac{1.53 - 1.89}{1.89} \times 100\% = -19\%$



Peak = 11.0715 V

Ripple = peak - valley

Ripple = 11.0715 - 9.5403

Ripple = 1.531 V

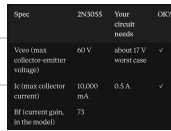
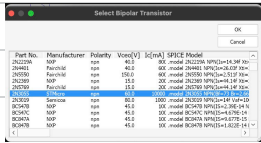
• Diode drop in simulation: $\frac{12.73 - 11.07}{2} = 0.83 \text{ V}$ per node,

close to 0.8 V assumption, which is why the peak matched within 0.5%.

• Load current: Average rail Voltage is about $\frac{11.07 + 9.54}{2} = 10.3 \text{ V}$, so $I = \frac{10.3}{20} = 0.515 \text{ A}$

• The ripple is 19% below my predicted because the formula assumes capacitor discharges for the whole half-cycle. The diodes actually turn back on sooner, so formula is conservative. Safe for design.

Choosing Transistor:

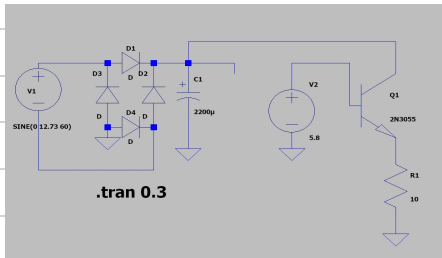
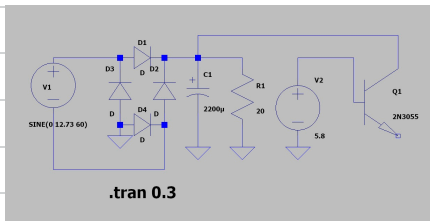


What is β ?

the base current divided by the gain:

$$I_B = I_C / \beta = 0.5 / 75 = 6.8 \text{ mA}$$

Two small current flows from 5.8 V reference into the base, which is easy for the reference to supply.



- With a real 7805, the load connects to the regulator's output, not the unregulated rail. All the load current has to flow through the regulator, so the load current also has to flow through Q_1 . Initially the 20 Ω hangs on the rail which causes a second, unwanted load.

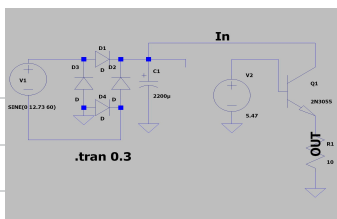
- Changed 20 Ω to 10 Ω after moving because $R = V/I \rightarrow R = 5.8 / 0.5 = 11.6 \Omega$, which gives my 500 mA test load. Its power is $P = I^2 R = 0.5^2 \cdot 10 = 2.5 \text{ W}$ which matches my 10 W resistors.

My expectations: Output: $V_{out} = V_{ref} - V_{BE} = 5.8 - V_{BE}$ • A V_{BE} near 0.8 V gives about 5 V.

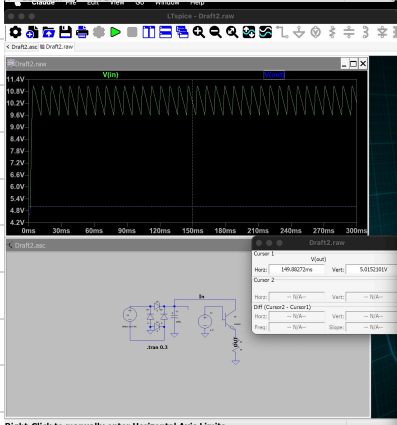
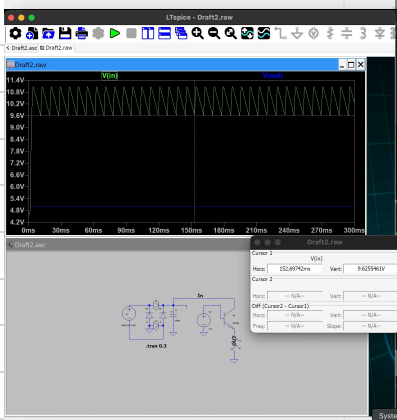
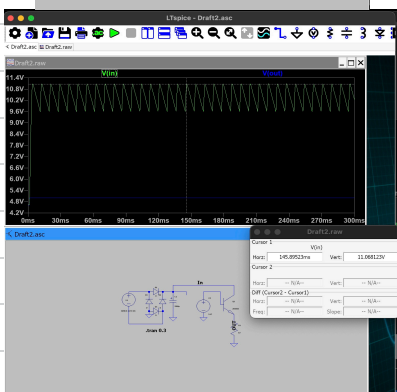
if it reads 4.9 V or 5.1 V, that's the model and I can adjust V_2 by 0.1 V (trimming).

Base current: $I_B = 0.5 / 75 = 6.8 \text{ mA}$, supplied by V_2

Rail: rail load is about 0.5 A again.



Measurement	Value
V(in) peak	11.0681 V
V(in) valley	9.6255 V
V(out)	5.0152 V



Right-Click to manually enter Horizontal Axis Limits

$$\text{Rail ripple} = 11.0681 - 9.6255 = 1.443 \text{ V}$$

$$\text{Rail Average} = \frac{11.0681 + 9.6255}{2} = 10.347 \text{ V}$$

$$\text{output error} = \frac{5.0152 - 5.00}{5.00} \times 100\% = 0.30\%$$

$$\text{load current} = \frac{5.0152}{10} = 0.5015 \text{ A}$$

$$\text{Regulator heat: } P = (10.347 - 5.0152) \times 0.5015 = 2.67 \text{ W}$$

Compared to hand Calculations:

Quantity	Calculated	Simulated	% error
Ripple	1.89 V	1.44 V	-24%
Regulator heat	2.59 W	2.67 W	+3.2%

V1 requirements table, simulation column			
Requirement	Target	Simulated	Pass?
Output voltage	5.0 V ±2%	5.015 V (+0.3%)	Pass
Dropout margin	valley > 7 V	9.63 V (+2.63 V margin)	Pass
Load	500 mA	501 mA	Pass
Output ripple < 50 mV		not valid in this model	Measure on the real circuit

